Appendix: Integration

The Riemann integral, multiple and improper integrals

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A.1 Definition of the Riemann integral

Motivation. The Riemann integral, approximation by continuous functions, and Lebesgue's criterion (integrable iff the discontinuities have measure zero).

Let f be bounded and real on $[a,b]$. A **partition** P is $a = x_0 < x_1 < \cdots < x_N = b$, with $I_j = [x_{j-1}, x_j]$, $|I_j| = x_j - x_{j-1}$; the upper and lower sums are

$$\mathcal{U}(P,f) = \sum_{j=1}^{N} \Big[\sup_{x\in I_j} f(x)\Big]|I_j|, \qquad \mathcal{L}(P,f) = \sum_{j=1}^{N} \Big[\inf_{x\in I_j} f(x)\Big]|I_j|.$$

$P' \supset P$ is a **refinement** of P, and then

$$\mathcal{L}(P,f) \le \mathcal{L}(P',f) \le \mathcal{U}(P',f) \le \mathcal{U}(P,f).$$

(splitting I_j does not increase the sups). With the common refinement $P_1 \cup P_2$, every lower sum is $\le$ every upper sum, so $\mathcal{L} = \sup_P \mathcal{L}(P,f) \le \mathcal{U} = \inf_P \mathcal{U}(P,f)$.

Definition

f is **(Riemann) integrable** if for every $\epsilon > 0$ there is a partition P with $\mathcal{U}(P,f) - \mathcal{L}(P,f) < \epsilon$. Then $\mathcal{U} = \mathcal{L}$, and this common value is $\int_a^b f(x)\,dx$. For complex $f = u + iv$, f is integrable if u and v are, and $\int f = \int u + i\int v$.

Equivalently, with Riemann sums $\mathcal{R}(P,f) = \sum f(x_j^*)|I_j|$, f is integrable iff $\mathcal{R}(P,f)$ converges as the mesh $\max|I_j| \to 0$. Continuous f are integrable: for small mesh $\sup_{I_j} f - \inf_{I_j} f < \epsilon$, so $\mathcal{U} - \mathcal{L} \le \epsilon(b-a)$.

A.1.1 Basic properties

Proposition 1.1

If f and g are integrable on $[a,b]$, then:
(i) $f+g$ is integrable, and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.
(ii) If $c \in \mathbb{C}$, then cf is integrable and $\int_a^b cf = c\int_a^b f$.
(iii) If f, g are real-valued and $f \le g$, then $\int_a^b f \le \int_a^b g$.
(iv) If $c \in [a,b]$, then f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$.

Proof. (i) Let f, g be real. $\mathcal{U}(P, f+g) \le \mathcal{U}(P,f) + \mathcal{U}(P,g)$ and $\mathcal{L}(P, f+g) \ge \mathcal{L}(P,f) + \mathcal{L}(P,g)$. With P_1, P_2 giving $\mathcal{U} - \mathcal{L} < \epsilon$ for f, g and a common refinement P_0, $\mathcal{U}(P_0, f+g) - \mathcal{L}(P_0, f+g) < 2\epsilon$, and the integral I satisfies

$$I \le \mathcal{U}(P_0, f+g) \le \mathcal{U}(P_0,f) + \mathcal{U}(P_0,g) \le \int f + \int g + 2\epsilon,$$

and likewise $I \ge \int f + \int g - 2\epsilon$.

(ii) For $c \ge 0$, $\mathcal{U}(P, cf) = c\,\mathcal{U}(P,f)$, $\mathcal{L}(P,cf) = c\,\mathcal{L}(P,f)$; $\mathcal{U}(P,-f) = -\mathcal{L}(P,f)$, $\mathcal{L}(P,-f) = -\mathcal{U}(P,f)$; complex c by (i).

(iii) $g - f \ge 0$ gives $\mathcal{L}(P, g-f) \ge 0$, so $\int(g-f) \ge 0$; apply (i) and (ii).

(iv) Take partitions containing c; then $\mathcal{U}(P,f) = \mathcal{U}(P',f) + \mathcal{U}(P'',f)$ for the induced partitions of $[a,c]$, $[c,b]$, likewise $\mathcal{L}$, so $\mathcal{U} - \mathcal{L} < \epsilon$ on each piece and the integrals add. ∎

Consequently $|f|$ is integrable with $\left|\int f\right| \le \int |f|$ (Lemma 1.2 with $\varphi(x) = |x|$; for complex f, $\varphi(u,v)$ on the bounded range, and $\int f = e^{i\theta}|\int f|$ gives $|\int f| = \int \mathrm{Re}(e^{-i\theta} f) \le \int |f|$), and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$ is integrable.

Lemma 1.2

If f is real-valued and integrable on $[a,b]$, and φ is a real-valued continuous function on $\mathbb{R}$, then $\varphi\circ f$ is integrable on $[a,b]$.

Proof. Let $|f|\le M$; by uniform continuity on $[-M,M]$ choose $\delta<\epsilon$ with $|\varphi(y_1)-\varphi(y_2)|<\epsilon$ for $|y_1-y_2|<\delta$, and P with $\mathcal{U}(P,f)-\mathcal{L}(P,f)<\delta^2$. Let Λ be the j with $\sup_{I_j} f-\inf_{I_j} f<\delta$ (there $\varphi\circ f$ oscillates $\le\epsilon$) and Λ' the rest. Then

$$\delta\sum_{j\in\Lambda'}|I_j|\le\sum_{j\in\Lambda'}\Big(\sup_{I_j}f-\inf_{I_j}f\Big)|I_j|<\delta^2,$$

so $\sum_{\Lambda'}|I_j|<\delta$, and with $B=\sup_{[-M,M]}|\varphi|$,

$$\mathcal{U}(P,\varphi\circ f)-\mathcal{L}(P,\varphi\circ f)\le\epsilon(b-a)+2B\delta<\epsilon(b-a+2B).$$

■

Proposition 1.3

A bounded monotonic function f on $[a,b]$ is integrable.

Proof. Take $a=0$, $b=1$, f increasing, and $x_j=j/N$:

$$\mathcal{U}(P_N,f)-\mathcal{L}(P_N,f)=\sum_{j=1}^{N}\frac{1}{N}[f(x_j)-f(x_{j-1})]=\frac{f(1)-f(0)}{N}\to 0.$$

■

Proposition 1.4

Let f be a bounded function on $[a,b]$. If $c\in(a,b)$, and for all small $\delta>0$ the function f is integrable on $[a,c-\delta]$ and $[c+\delta,b]$, then f is integrable on $[a,b]$.

Proof. Let $|f|\le M$. Choose $4\delta M\le\epsilon/3$ and partitions P_1 of $[a,c-\delta]$, P_2 of $[c+\delta,b]$ with $\mathcal{U}-\mathcal{L}<\epsilon/3$; $P=P_1\cup P_2$ has $[c-\delta,c+\delta]$ as a subinterval, so

$$\mathcal{U}(P,f)-\mathcal{L}(P,f)\le(\mathcal{U}(P_1,f)-\mathcal{L}(P_1,f))+2M\cdot 2\delta+(\mathcal{U}(P_2,f)-\mathcal{L}(P_2,f))<\epsilon.$$

■

By induction, bounded functions continuous except at finitely many points are integrable.

A function on the circle (2π-periodic on $\mathbb{R}$) is **integrable** if integrable on $[-\pi,\pi]$.

Lemma 1.5

Suppose f is integrable on the circle and $|f|\le B$. Then there is a sequence $\{f_k\}$ of continuous functions on the circle with

$$\sup_{x\in[-\pi,\pi]}|f_k(x)|\le B\ \text{ for all } k,\qquad\int_{-\pi}^{\pi}|f(x)-f_k(x)|\,dx\to 0\ \text{ as } k\to\infty.$$

Proof. Take f real (complex: treat parts separately, then truncate radially to B, a 1-Lipschitz operation).

Step Step function. Choose $-\pi = x_0 < \cdots < x_N = \pi$ with $\mathcal{U} - \mathcal{L} < \epsilon$ and $f^* = \sup_{[x_{j-1},x_j]} f$ on $[x_{j-1}, x_j)$. Then $|f^*| \le B$ and

$$\int_{-\pi}^{\pi} |f^* - f| = \int_{-\pi}^{\pi} (f^* - f) \le \mathcal{U} - \mathcal{L} < \epsilon.$$

Step Smoothing the jumps. For small δ, let $\tilde{f} = f^*$ away from the x_j, linear on $[x_j - \delta, x_j + \delta]$ $(1 \le j \le N-1)$ matching $f^*(x_j \pm \delta)$, and linear on $[-\pi, -\pi + \delta]$ and $[\pi - \delta, \pi]$ with $\tilde{f}(\pm\pi) = 0$. Then $\tilde{f}$ is continuous, 2π-periodic, $|\tilde{f}| \le B$, and differs from f^* on N intervals of length $\le 2\delta$: $\int |f^* - \tilde{f}| \le 4BN\delta < \epsilon$.

So $\int |f - \tilde{f}| < 2\epsilon$; take $\epsilon = 1/k$. ■

A.1.2 Sets of measure zero and discontinuities of integrable functions

$E \subset \mathbb{R}$ has **measure** 0 if for every $\epsilon > 0$ countably many open intervals I_k cover E with $\sum |I_k| < \epsilon$. Countable sets have measure 0 (lengths $\epsilon 2^{-k}$); a compact set of measure 0 is covered by finitely many I_k.

Lemma 1.6

The union of countably many sets of measure 0 has measure 0.

Proof. Cover E_i by $\{I_{k,i}\}_k$ with $\sum_k |I_{k,i}| < \epsilon/2^i$; together they cover $\bigcup_i E_i$ with total length $< \epsilon$. ■

Theorem 1.7 (Lebesgue)

A bounded function f on $[a, b]$ is integrable if and only if its set of discontinuities has measure 0.

Let $J = [a, b]$, $I(c, r) = (c - r, c + r)$, $\operatorname{osc}(f, c, r) = \sup_{x,y \in J \cap I(c,r)} |f(x) - f(y)|$ and $\operatorname{osc}(f, c) = \lim_{r \to 0} \operatorname{osc}(f, c, r)$; f is continuous at c iff $\operatorname{osc}(f, c) = 0$. With $A_\epsilon = \{c \in J : \operatorname{osc}(f, c) \ge \epsilon\}$, the discontinuity set is $D = \bigcup_n A_{1/n}$.

Lemma 1.8

For $\epsilon > 0$, the set A_ϵ is closed, hence compact.

Proof. If $c_n \in A_\epsilon$, $c_n \to c \notin A_\epsilon$, write $\operatorname{osc}(f, c) = \epsilon - \delta$, $\delta > 0$, choose r with $\operatorname{osc}(f, c, r) < \epsilon - \delta/2$ and $|c_n - c| < r/2$. As $I(c_n, r/2) \subset I(c, r)$, $\operatorname{osc}(f, c_n) < \epsilon$: a contradiction. ■

Proof of Theorem 1.7. Let $|f| \le M$.

Step Measure zero ⇒ integrable. $A_\epsilon \subset D$ is compact of measure 0, so finitely many open $I_1, \ldots, I_N$ of total length $< \epsilon$ cover it. Each point z of the compact rest of J has an open interval F_z on which f oscillates $\le \epsilon$; take a finite subcover $I_{N+1}, \ldots, I_{N'}$. The endpoints give a partition P whose subintervals lie in the closure of some I_k, $k \le N$ (contributing $\le 2M$ times length), or in some I_k, $k > N$ (oscillation $\le \epsilon$). So

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) \le 2M \sum_{j=1}^{N} |I_j| + \epsilon(b - a) \le (2M + b - a)\epsilon.$$

Step Integrable ⇒ measure zero. By Lemma 1.6 show each $A_{1/n}$ has measure 0. Take P with $\mathcal{U}(P,f)-\mathcal{L}(P,f)<\epsilon/n$. If $A_{1/n}$ meets $I_j=(x_{j-1},x_j)$, then $\sup_{I_j} f-\inf_{I_j} f\geq 1/n$, so

$$\frac{1}{n}\sum_{j:\, I_j\cap A_{1/n}\neq\emptyset}|I_j|\leq \mathcal{U}(P,f)-\mathcal{L}(P,f)<\frac{\epsilon}{n}.$$

Enlarging these intervals slightly and adding tiny intervals around the x_j covers $A_{1/n}$ with total length $\leq 2\epsilon$. ■

(So fg is integrable: its discontinuities lie in those of f and g.)

A.2 Multiple integrals

Motivation. Integrals over rectangles in $\mathbb{R}^d$: Fubini, change of variables with $|\det Dg|$, and spherical coordinates.

A.2.1 The Riemann integral in $\mathbb{R}^d$

A closed rectangle is $R=[a_1,b_1]\times\cdots\times[a_d,b_d]$; partitions P_j of $[a_j,b_j]$ give a partition P of R into $S=S_1\times\cdots\times S_d$, $|S|=|S_1|\cdots|S_d|$. Upper and lower sums

$$\mathcal{U}(P,f)=\sum_S\Big[\sup_S f\Big]|S|,\qquad \mathcal{L}(P,f)=\sum_S\Big[\inf_S f\Big]|S|,$$

refinements, integrability and $\int_R f$ are as in one dimension; continuous f are integrable. For f continuous on a closed ball B (or a set with boundary of measure zero), $g=f\mathbb{1}_B$ is integrable on any rectangle $R\supset B$ (discontinuities on ∂B), and $\int_B f=\int_R g$ is independent of R.

A.2.2 Repeated integrals

Theorem 2.1 (Fubini, continuous case)

Let f be continuous on a closed rectangle $R=R_1\times R_2\subset\mathbb{R}^d$, $R_i\subset\mathbb{R}^{d_i}$, $d=d_1+d_2$. Writing $x=(x_1,x_2)$, the function $F(x_1)=\int_{R_2} f(x_1,x_2)\,dx_2$ is continuous on R_1, and

$$\int_R f(x)\,dx=\int_{R_1}\Big(\int_{R_2} f(x_1,x_2)\,dx_2\Big)dx_1.$$

Proof.

Step Continuity. $|F(x_1)-F(x_1')|\leq|R_2|\sup_{x_2}|f(x_1,x_2)-f(x_1',x_2)|$, small for x_1' near x_1 by uniform continuity.

Step Identity. For partitions P_1,P_2 of R_1,R_2 with subrectangles S,T,

$$\sup_{S\times T} f\geq\sup_{x_1\in S}\Big(\sup_{x_2\in T} f(x_1,x_2)\Big),\qquad \inf_{S\times T} f\leq\inf_{x_1\in S}\Big(\inf_{x_2\in T} f(x_1,x_2)\Big).$$

For each x_1,

$$\sum_T\Big[\inf_{x_2\in T} f(x_1,x_2)\Big]|T|\leq F(x_1)\leq\sum_T\Big[\sup_{x_2\in T} f(x_1,x_2)\Big]|T|.$$

Taking $\sup_{x_1\in S}$ and using the first inequality,

$$\mathcal{U}(P_1,F)\leq\sum_S\sum_T\Big[\sup_{S\times T} f\Big]|T||S|=\mathcal{U}(P,f),$$

and likewise $\mathcal{L}(P,f) \le \mathcal{L}(P_1,F)$. As $\mathcal{U}(P,f) - \mathcal{L}(P,f)$ can be made small, $\int_{R_1} F = \int_R f$. ■

By induction $\int_R f = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f\, dx_d \cdots dx_1$, in any order.

A.2.3 The change of variables formula

$g : A \to B$ between compact sets is a C^1 **diffeomorphism** if it is a bijection extending to a C^1 map near A, with inverse extending to a C^1 map near B, and $\det Dg \neq 0$ on A.

Theorem 2.2 (change of variables)

Suppose A and B are compact subsets of $\mathbb{R}^d$ and $g : A \to B$ is a C^1 diffeomorphism. If f is continuous on B, then

$$\int_{g(A)} f(x)\, dx = \int_A f(g(y))\, |\det(Dg)(y)|\, dy.$$

Proof. [Supplement] Assume (implicitly) that A, B have boundaries of measure zero, and write $\mathrm{vol}(E) = \int_E 1$.

Step Linear maps. For invertible linear L and a rectangle R, $\mathrm{vol}(L(R)) = |\det L|\,\mathrm{vol}(R)$. Both sides are multiplicative in L ($\mathrm{vol}(L(E)) = c_L \mathrm{vol}(E)$, approximating E by rectangles), so check elementary matrices: permutations and scalings map rectangles to rectangles with the right volume, and for a shear $x_1 \mapsto x_1 + t x_2$ Fubini (Theorem 2.1) shows each x_1-slice is translated, so volume is preserved ($\det = 1$).

Step Local estimate. Fix $\epsilon \in (0,1)$. By uniform continuity of Dg and $(Dg)^{-1}$ on A there is $\eta > 0$ such that for every cube $Q \subset A$ of side $< \eta$ and centre a,

$$g(a) + Dg(a)\big((1-\epsilon)(Q-a)\big) \subset g(Q) \subset g(a) + Dg(a)\big((1+\epsilon)(Q-a)\big). \tag{A.1}$$

Indeed $h = Dg(a)^{-1}(g - g(a))$ satisfies $|h(y) - (y-a)| \le \epsilon' |y-a|$ (mean value inequality, ϵ' uniform in a), so h maps $Q-a$ into $(1+\epsilon)(Q-a)$, and, being a homeomorphism moving ∂Q by less than the gap, covers $(1-\epsilon)(Q-a)$. By the linear case and (A.1),

$$(1-\epsilon)^d |\det Dg(a)|\, |Q| \le \mathrm{vol}(g(Q)) \le (1+\epsilon)^d |\det Dg(a)|\, |Q|.$$

Step Summation. Take $f \ge 0$ (general f by linearity). Partition a rectangle containing A into cubes Q_k of side $< \eta$; let $\mathcal{K}$ be those inside A. The others meet ∂A, and they and their images (g Lipschitz) have volume $\to 0$ as $\eta \to 0$. The $g(Q_k)$ overlap only on null boundaries, so

$$\int_{g(A)} f = \sum_{k \in \mathcal{K}} \int_{g(Q_k)} f + o(1) = \sum_{k \in \mathcal{K}} f(g(a_k))\, \mathrm{vol}(g(Q_k)) + o(1),$$

with a_k the centre of Q_k (uniform continuity of f). By the local estimate this is within a factor $(1 \pm \epsilon)^d$ of the Riemann sum $\sum_k f(g(a_k)) |\det Dg(a_k)| |Q_k|$ for $\int_A f(g(y)) |\det Dg(y)| dy$. Let $\eta \to 0$, then $\epsilon \to 0$. ■

$|\det Dg|$ is the new volume element: $|L(R)| = |\det L|\, |R|$ for linear L.

A.2.4 Spherical coordinates

Spherical coordinates in $\mathbb{R}^d$ are $x = g(r, \theta_1, \dots, \theta_{d-1})$ with

$$\begin{aligned} x_1 &= r\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{d-2}\cos\theta_{d-1},\\ x_2 &= r\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{d-2}\sin\theta_{d-1},\\ &\vdots\\ x_{d-1} &= r\sin\theta_1\sin\theta_2, \qquad x_d = r\cos\theta_1, \end{aligned}$$

with $0 \le \theta_i \le \pi$ $(i \le d-2)$, $0 \le \theta_{d-1} \le 2\pi$, and Jacobian $r^{d-1}\sin^{d-2}\theta_1 \sin^{d-3}\theta_2 \cdots \sin\theta_{d-2}$. Every $x \neq 0$ is uniquely $r\gamma$, $r > 0$, $\gamma \in S^{d-1}$. Define

$$\int_{S^{d-1}} f(\gamma)\, d\sigma(\gamma) = \int_0^\pi \cdots \int_0^\pi \int_0^{2\pi} f(g(1,\theta)) \sin^{d-2}\theta_1 \cdots \sin\theta_{d-2}\, d\theta_{d-1}\cdots d\theta_1.$$

Change of variables gives, for the ball $B(0,N)$,

$$\int_{B(0,N)} f(x)\, dx = \int_{S^{d-1}} \int_0^N f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma). \tag{A.2}$$

(g fails to be a diffeomorphism on a null set: apply Theorem 2.2 on $\{\delta \le r,\ \delta \le \theta_i \le \pi - \delta\}$ and let $\delta \to 0$.) The unit sphere has area $\omega_d = \int_{S^{d-1}} d\sigma$.

For $A(R_1, R_2) = \{R_1 \le |x| \le R_2\}$ and $\lambda \in \mathbb{R}$,

$$\int_{A(R_1,R_2)} |x|^\lambda\, dx = \omega_d \int_{R_1}^{R_2} r^{\lambda+d-1} dr = \begin{cases} \dfrac{\omega_d}{\lambda+d}\left(R_2^{\lambda+d} - R_1^{\lambda+d}\right), & \lambda \neq -d,\\ \omega_d \log(R_2/R_1), & \lambda = -d. \end{cases}$$

So $|x|^\lambda$ is integrable near 0 iff $\lambda > -d$, and near ∞ iff $\lambda < -d$.

A.3 Improper integrals. Integration over $\mathbb{R}^d$

Motivation. Over all of $\mathbb{R}^d$, integrals of continuous functions of moderate decrease are limits over growing cubes or balls; Fubini and polar coordinates extend.

A.3.1 Integration of functions of moderate decrease

Let $Q_N = \{|x_j| \le N/2\}$; for continuous f, $\int_{\mathbb{R}^d} f = \lim_N \int_{Q_N} f$ if it exists. f is of **moderate decrease** if

$$|f(x)| \le \frac{A}{1+|x|^{d+1}} \quad \text{for some } A > 0.$$

(For $d = 1$ as in Chapter 5; e.g. $\mathcal{P}_y(x) = \frac{1}{\pi}\frac{y}{x^2+y^2}$.)

Then the limit exists: with $I_N = \int_{Q_N} f$ and $M > N$, $Q_M - Q_N \subset A(aN, bM)$, $a = 1/2$, $b = \sqrt{d}/2$ (as $\{|x| < N/2\} \subset Q_N \subset \{|x| \le \sqrt{d}N/2\}$), so by the annulus formula with $\lambda = -d-1$

$$|I_M - I_N| \le A \int_{aN \le |x| \le bM} |x|^{-d-1} dx = A\omega_d\Big(\frac{1}{aN} - \frac{1}{bM}\Big) \le \frac{C}{N}.$$

So $\{I_N\}$ is Cauchy; over balls the limit is the same, as $|\int_{Q_N} f - \int_{B_{N/2}} f| \le C/N$.

The integral is linear and invariant under translations and rotations, and $\int f(\delta x)\, dx = \delta^{-d}\int f$: let $N \to \infty$ in the statements on cubes or balls (Theorem 2.2, $|\det| = 1$ or δ^d), translates of Q_N differing from Q_N where $|f|$ contributes $O(1/N)$.

A.3.2 Repeated integrals

Theorem 3.1

Suppose f is continuous on $\mathbb{R}^2$ and of moderate decrease. Then $F(x_1) = \int_{\mathbb{R}} f(x_1, x_2)\, dx_2$ is of moderate decrease on $\mathbb{R}$, and

$$\int_{\mathbb{R}^2} f(x)\, dx = \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} f(x_1, x_2)\, dx_2 \Big) dx_1.$$

Proof.

Step Decrease of F. With $|f| \le A/(1+|x|^3)$, split at $|x_2| = |x_1|$: on $|x_2| \le |x_1|$ the integrand is $\le A/(1+|x_1|^3)$, giving $\le 2A|x_1|/(1+|x_1|^3) \le A'/(1+|x_1|^2)$; on $|x_2| \ge |x_1|$ it is $\le A/(1+|x_2|^3)$, giving $\le A'''/(1+|x_1|^2)$. F is continuous: integrals over $|x_2| \le N$ are (Theorem 2.1) and converge uniformly, the tails being $\le \int_{|x_2|>N} A/(1+|x_2|^3)$.

Step Identity. Choose N with $\big|\int_{\mathbb{R}^2} f - \int_{Q_N} f\big| < \epsilon$, $\int_{|x_1|>N/2} |F| < \epsilon$ and $\int_{|x_1|\le N/2} \int_{|x_2|>N/2} |f| < \epsilon$ (the inner integral is $\le \min\big(\frac{C}{1+|x_1|^2}, \frac{C}{N}\big)$). By Theorem 2.1,

$$\int_{Q_N} f = \int_{|x_1|\le N/2} \int_{|x_2|\le N/2} f\, dx_2\, dx_1,$$

within ϵ of $\int_{|x_1|\le N/2} F$, itself within ϵ of $\int_{\mathbb{R}} F$. So $\big|\int_{\mathbb{R}^2} f - \int_{\mathbb{R}} F\big| < 3\epsilon$. ∎

Likewise in $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$.

A.3.3 Spherical coordinates

For f of moderate decrease and fixed $\gamma \in S^{d-1}$,

$$|f(r\gamma) r^{d-1}| \le \frac{A r^{d-1}}{1 + r^{d+1}} \le \frac{B}{1+r^2},$$

so $r \mapsto f(r\gamma) r^{d-1}$ is of moderate decrease, and $N \to \infty$ in (A.2) (uniformly in γ) gives

$$\int_{\mathbb{R}^d} f(x)\, dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

With rotation invariance of $\int_{\mathbb{R}^d}$, this gives rotation invariance of $d\sigma$:

$$\int_{S^{d-1}} f(R(\gamma))\, d\sigma(\gamma) = \int_{S^{d-1}} f(\gamma)\, d\sigma(\gamma) \tag{A.3}$$

for continuous f on S^{d-1} (apply the formula to $F(x) = f(x/|x|)\varphi(|x|)$, $\varphi \ge 0$ continuous with compact support in $(0,\infty)$ and $\int_0^\infty \varphi(r) r^{d-1} dr = 1$).